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Study on Non-Equilibrium Diffusion Characteristics of Supercritical Condensate Gas and Dry Gas

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Abstract

Condensate gas reservoirs development usually adopts the dry gas injection to enhance condensate recovery. Conventional understanding suggests complete miscibility between condensate gas and injected dry gas without biphasic interface formation. However, field observations from the Yaha condensate gas reservoir challenge this assumption; during the injection process, the interface between the two gases evolves into a transition zone. At present, there is little research to describe this phenomenon mathematically.

The study first designed experimental procedures for condensate gas and dry gas diffusion in PVT cells based on the pressure-temperature conditions, fluid composition of the Yaha condensate gas reservoir. These experiments were conducted to investigate interfacial characteristics during the development process. Subsequently, through systematic observation of phase behavior features including interface variations, a theoretical diffusion model was established to quantitatively characterize the dynamic evolution of three distinct zones: the injected dry gas slug, transition zone, and original condensate gas region. This model effectively captures the transient interactions and mass transfer processes between the injected gas and reservoir fluids.

The visual PVT experiment demonstrates the vertical diffusion process between condensate gas and dry gas at a non-equilibrium state, providing the experimental basis for the theoretical model. The non-equilibrium diffusion model could calculate the diffusion coefficient of each component and fluid density variation at different positions in the PVT cell during the diffusion process. The results indicate that the density of condensate gas is much higher than that of dry gas and the molecular diffusion coefficient of each component in condensate gas is lower than that in dry gas by roughly 1–2 orders of magnitude. Besides, the diffusion coefficients of components are inversely proportional to their molecular weights and tend to decrease more rapidly at the initial stage. During the diffusion process, the density difference between dry gas and condensate gas gradually decreases along with the thickening of the transition zone and reaches a minimum value at the equilibrium state.

In dynamic processes of hydrocarbon reservoir development, non-equilibrium states should be recognized as universally prevalent, while complete equilibrium conditions prove fundamentally inadequate. This finding necessitates re-evaluation of phase behavior models for gas injection processes in condensate reservoirs, provides another solution for accurately characterizing gas injection phase behavior and displacement features, thereby enabling more realistic evaluation of enhanced recovery mechanisms.

Introduction

Deep condensate gas reservoirs in western China often exhibit supercritical properties (e.g., high density, low viscosity) under reservoir conditions. Dry gas injection for pressure maintenance induces "gravity override," reducing cyclic injection efficiency over time.

The condensate gas with high condensate content in the deep formation of western China is often characterized by supercritical properties, such as high density and low viscosity under reservoir conditions. The "gravity override" appears during the dry cyclic gas injection stage to maintain reservoir pressure. The efficiency of cyclic gas injection weakens with time, and the development performance cannot meet the expectations. The formation demonstrates different gas-condensate ratios and gravity differentiation of hydrocarbon components. Therefore, the determination of the molecular diffusion coefficient of the supercritical condensate gas system has important theoretical significance for accurately describing the development characteristics of a condensate gas reservoir.

The whole system lies in a non-equilibrium state before the two gases are thoroughly mixed. Most of the theoretical models of non-equilibrium diffusion are carried out at the PVT cell scale and mainly address gas diffusion in the liquid phase. Different components' diffusion coefficients can be calculated precisely by analyzing the composition and concentration of fluid samples at different times and positions. Alternatively, the diffusion coefficient can be obtained indirectly by fitting the system's pressure change curve.

Shi et al. (2023) proposed a phase determination method for non-equilibrium processes in oil-gas systems of gas storage reservoirs. Bayestehparvin et al. (2022) put forward that non-equilibrium phase behavior plays a role in solvent-aided processes. Riazi (1966) first established a non-equilibrium gas-liquid diffusion model in a PVT cell, considering that the change of pressure and interface position over time depends on the diffusion coefficient of each phase. Sachs (1998) presented a numerical solution to the nonlinear model equation of methane in liquid water at high pressure, taking into account the relationship between diffusion coefficient and concentration. Zhang et al. (2000) combined the diffusion equation with the gas material balance equation based on Riazi and tested the diffusion coefficient of the gas in heavy oil. Civan et al. (2002) improved the non-equilibrium diffusion model, accordingly, taking account of the resistance of gas-liquid interface conditions to gas dissolution abilities in liquid, and measured the diffusion coefficient and interfacial mass transfer coefficient. Tharanivasan et al. (2004) studied the influence of equilibrium, pseudo-equilibrium, and non-equilibrium boundary conditions on the diffusion coefficient and gave the analytical solutions to the diffusion equation under those conditions.

After saturating the porous media in the liquid phase, Islas et al. (2004) established the gas-liquid diffusion model in the porous media and studied the method for testing effective diffusion coefficients in the porous media. Guo et al. (2009) established a mathematical model for the oil and gas diffusion process with multi-components and studied the influence of gas concentration on diffusion coefficient. Unatrakarn et al. (2011) estimated the diffusion coefficient of the gas in the oil phase and porous media by adjusting the gas diffusion coefficient to fit the pressure-time relationship. Etminan et al. (2013) proposed the general analytical solution of the interface model, which can better explain how the interface resistance affects the diffusion process.

Kavousi et al. (2013) established the mathematical model of CO₂ diffusion in heavy oil and studied solubility and diffusion coefficient under static and dynamic conditions. Jooybari et al. (2014) believed that Robin boundary conditions have little influence on the gas diffusion process. Roman et al. (2015)

established a gas phase pressure attenuation model and directly estimated the diffusion coefficient using a graphical method. In conclusion, the references dealt with diffusion problems in gas-liquid phases and barely covered what occurred in supercritical gas phases. Besides, the existing models are inadequate to describe the diffusion in supercritical conditions. [Ma et al. \(2024\)](#) developed a non-equilibrium gas injection method suitable for fault-controlled fracture-cavity carbonate condensate gas reservoirs to enhance condensate gas recovery. [Jiang et al. \(2021\)](#) investigated the ternary development mechanism of cyclic gas injection in condensate gas reservoirs and novel technologies for improving condensate oil recovery, elucidating the dominant driving forces under different reservoir conditions.

Based on visual experiments on the vertical diffusion behavior of supercritical condensate gas and dry gas and considering the diffusion transition zone, we improved the existing gas-liquid diffusion model and established a non-equilibrium diffusion model for condensate gas and dry gas. The calculation of the diffusion coefficient considers the dynamic diffusion characteristics over time. Besides, the gas diffusion equation is combined with the actual gas equation of state. Considering the unique physical properties under high pressure, we calculated the deviation coefficient of supercritical condensate gas by the DAK method, and obtained the pressure by the PR equation of state. In the end, the dynamic diffusion coefficients of each component in condensate gas were accurately calculated and compared with those of conventional dry gas.

Condensate Gas-Dry Gas Diffusion Experiment

The experiment was carried out using JEFFRI Visualized Formation Fluid Analyzer to test the diffusion characteristics of dry gas injected at the top of condensate gas formation.

In this section, two condensate gas samples were chosen, one with less oil content and one with high oil content. Details of the samples can be found in [Table 1](#). [Figure 1](#) shows the pressure-temperature (P-T) phase diagram of the condensate gas sample used in PVT tests.

Table 1—Condensate gas and dry gas composition

Component	Condensate gas (%)		Dry gas (%)
	Low oil content	High oil content	
CO ₂	2.12	0.59	0
N ₂	3.09	3.06	0.531
C ₁	83.19	80.02	96.416
C ₂	4.64	8.17	1.789
C ₃	1.93	2.57	0.834
iC ₄	0.35	0.53	0.148
nC ₄	0.83	0.65	0.166
iC ₅	0.33	0.26	0.058
nC ₅	0.51	0.2	0.04
C ₆	0.75	0.22	0.019
C ₇₊	2.26	3.73	0
Saturation pressure (MPa)	44.58	43.06	-

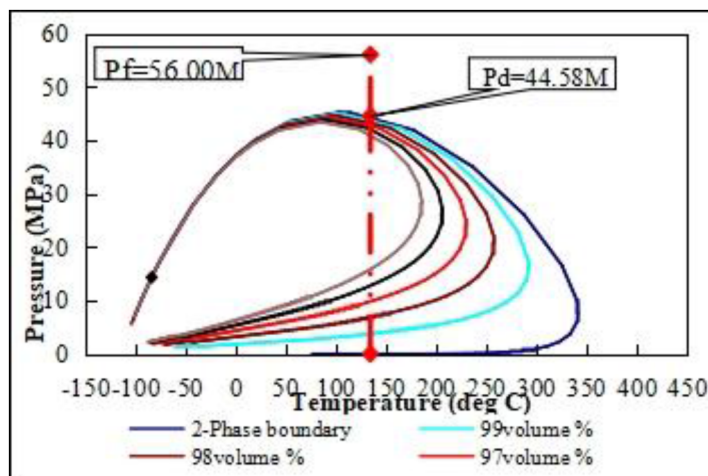


Figure 1—Fluid P-T phase diagram of formation condensate gas (Pf: formation pressure, Pd: dew point pressure)

The experimental procedures are as follows: 1) Transfer the condensate gas sample into the PVT cell to reach a stable state under formation temperature and pressure conditions. 2) Inject the dry gas slowly from the top of the condensate gas. Note that this process requires simultaneous operation of two pumps, one injection, and one extraction to maintain constant pressure with a slow rate to avoid turbulence generation. 3) When the volume ratio of dry gas and condensate gas reaches 1:1, stop the injection, close the valve to maintain constant temperature and pressure. 4) Take photos at a specific time interval and record relevant data.

Figure 2 and Figure 3 show the vertical diffusion phase behavior of dry gas and condensate gas with different oil contents (high for Figure 2 and low for Figure 3). The time interval between photos is 8 minutes, and it takes 176 and 120 minutes for the interfaces to vanish separately. It can be concluded that systems with lower oil content reach equilibrium faster.

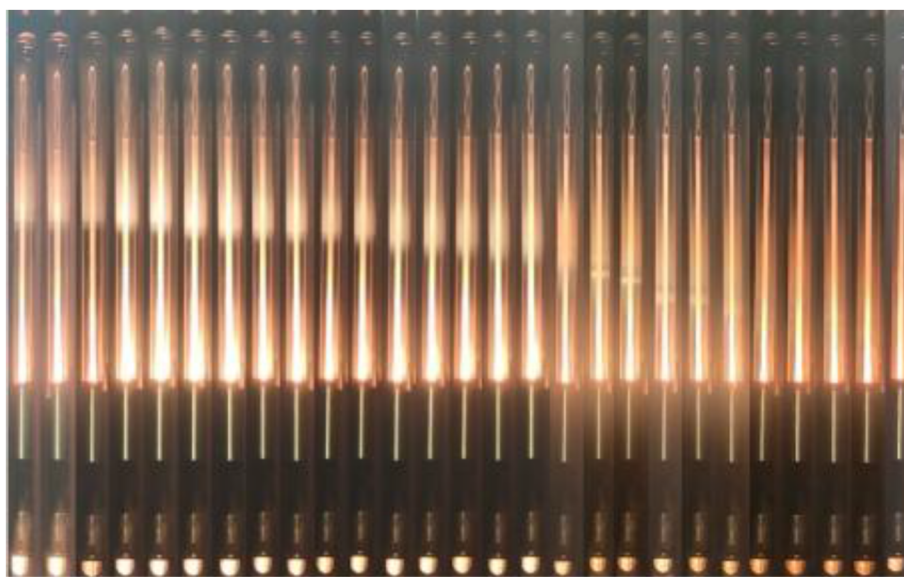


Figure 2—Sequential photos of the non-equilibrium diffusion process between supercritical condensate gas and dry gas (high oil content)



Figure 3—Sequential photos of the non-equilibrium diffusion process between supercritical condensate gas and dry gas (low oil content)

The change of phase behavior can be divided into two stages: 1) when the dry gas is injected, due to the density difference and the influence of the extraction and diffusion, the fog-like phenomenon appears in the PVT tube, and the interface is prominent and relatively stable. 2) Afterwards, the interface blurs and evolves into a transition zone which gradually grows with time; 3) In the end, the transition zone almost takes up all the PVT cells, and the interface disappears upon reaching equilibrium.

Table 2—Diffusion coefficients of different components in different phases

Component	Dry Gas(m ² /s)	Condensate Gas(m ² /s)
CO ₂	\	9.658E-07
N ₂	4.617E-05	8.498E-06
C ₁	1.258E-04	9.223E-05
C ₂	4.020E-05	7.842E-06
C ₃	2.060E-05	6.574E-07
iC ₄	7.804E-06	1.971E-07
nC ₄	5.565E-06	1.385E-07
iC ₅	1.988E-06	3.726E-08
nC ₅	2.471E-06	3.951E-08
C ₆	9.693E-07	4.146E-09

Theoretical analysis

Assumptions

Based on the phenomenon observed in the diffusion experiment, we developed a model of a PVT cell with a specified volume, as shown in Figure 4. It contains dry gas and condensate gas with specific compositions at a supercritical non-equilibrium state. Due to the two gases' significant density difference, a distinct interface is formed between condensate gas-dry gas. After that, molecular diffusion occurs under the action of the concentration difference, which reduces the composition difference and forms a transition zone similar to the miscible flooding.

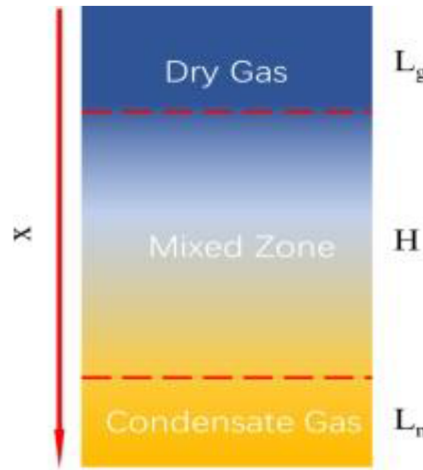


Figure 4—Schematic diagram of the physical model

The temperature and total volume remain constant during the whole diffusion process, while the pressure, density, composition of each phase, and diffusion coefficient change with time. Assuming that the initial height of the transition zone is 0, with the development of diffusion, three sections are present: the injected gas at the top, the transition zone in the middle, and the condensate gas at the bottom. The thickness of the condensate gas L_n and dry gas L_g gradually decreases, while the thickness of the transition zone H gradually increases.

Mathematical formulation

According to the continuity equation and Fick's second law, the following one-dimensional unsteady diffusion equation can be used to describe the molecular diffusion process (*Bird, R.B., 1960*):

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} \quad (1)$$

Where t is the diffusion time, x is the distance from the top of the PVT cell, and concentration c is a function of t and x .

The condensate phase:

$$\frac{\partial c_{ni}}{\partial t} = D_{ni} \frac{\partial^2 c_{ni}}{\partial x^2} \quad (2)$$

Initial conditions: at the beginning of diffusion, the initial concentration of component i in the condensate gas phase is given by:

$$c_{ni} = C_{ni} \quad L_g + H < x \leq L, t = 0 \quad (3)$$

Boundary conditions: During the diffusion, the condensate gas phase will not reach the upper boundary, and the lower part of the condensate gas phase does not participate in diffusion.

$$\frac{\partial c_{ni}}{\partial x} = 0, x = 0, t > 0 \quad (4)$$

$$\frac{\partial c_{ni}}{\partial x} = 0, x = L, t > 0 \quad (5)$$

The transition zone:

$$\frac{\partial c_{hi}}{\partial t} = D_{hi} \frac{\partial^2 c_{hi}}{\partial x^2} \quad (6)$$

Initial condition: at the beginning of diffusion, the concentration of component i in the transition zone is the same as that of component i in the dry gas phase.

$$c_{hi} = C_{hi} \quad L_g < x < L_g + H, t = 0 \quad (7)$$

Boundary conditions: the diffusion of the transition zone does not reach the bottom of the cell, and the upper boundary of the dry gas phase does not participate in the diffusion.

$$\frac{\partial c_{hi}}{\partial x} = 0, x = 0, t > 0 \quad (8)$$

$$\frac{\partial c_{hi}}{\partial x} = 0, x = L, t > 0 \quad (9)$$

Solution Methodology

For the condensate gas phase, a dimensionless concentration is defined firstly:

$$f = \frac{c_{ni}}{c_{ni}} \quad (10)$$

Then Equation (1) becomes:

$$\frac{\partial f}{\partial t} = D_{ni} \frac{\partial^2 f}{\partial x^2} \quad (11)$$

Set $f(t, x) = h(x) \cdot g(t)$, substitute in Equation (11) leads to:

$$h(x) \cdot g'(t) - D_{ni} h''(x) \cdot g(t) = 0 \quad (12)$$

Which equals:

$$\frac{h''(x)}{h(x)} = \frac{g'(t)}{D_{ni} g(t)} \quad (13)$$

Since the left side of the equation is a function of x and the right side is a function of t. To make equation (13) stand, the functions at both ends of the equation must equal to the constant a, then solving via separation of variables yields:

$$\frac{h''(x)}{h(x)} = \frac{g'(t)}{D_{ni} g(t)} = a \quad (14)$$

Therefore, the solution of the original equation can be transformed into the general solution of two linear differential equations:

$$\frac{h''(x)}{h(x)} = a \quad (15)$$

$$\frac{g'(t)}{D_{ni} g(t)} = a \quad (16)$$

The general solution of Equation (16) is:

$$g(t) = K \exp(a D_{ni} t) \quad (17)$$

In Equation (15), the constant a is a negative number and is not 0, assume $a = -\lambda^2$, then Equation (15) becomes:

$$h''(x) + \lambda^2 h(x) = 0 \quad (18)$$

The general solution of Equation (18) is:

$$h(x) = K_1 \sin \lambda x + K_2 \cos \lambda x \quad (19)$$

Equation (17) is rewritten as:

$$g(t) = K \exp(-\lambda^2 D_{ni} t) \quad (20)$$

Therefore, the general solution of Equation (11) is:

$$f(t, x) = (K_1 \sin \lambda x + K_2 \cos \lambda x) K \exp(-\lambda^2 D_{ni} t) = (A \sin \lambda x + B \cos \lambda x) \exp(-\lambda^2 D_{ni} t) \quad (21)$$

The first-order partial derivative is:

$$\frac{\partial f(t, x)}{\partial x} = (\lambda A \cos \lambda x - \lambda B \sin \lambda x) \exp(-\lambda^2 D_{ni} t) \quad (22)$$

Rewrite the boundary conditions and the initial conditions as:

Boundary conditions :

$$\frac{\partial f}{\partial x} = 0, x = 0, t > 0 \quad (23)$$

$$\frac{\partial f}{\partial x} = 0, x = L, t > 0 \quad (24)$$

Initial conditions:

$$f = 1, L_g + H < x < L, t = 0 \quad (25)$$

From the boundary conditions, we get:

$$A = 0 \quad (26)$$

$$\lambda = \frac{k\pi}{L} (k = 1, 2, \dots, \infty) \quad (27)$$

Therefore:

$$f(t, x) = \sum_{k=1}^{\infty} B \exp\left(-\frac{k^2 \pi^2}{L^2} D_{ni} t\right) \cos \frac{k\pi}{L} x \quad (28)$$

From the initial conditions:

$$f(0, x) = \sum_{k=1}^{\infty} B \cos \frac{k\pi}{L} x = 1 \quad (29)$$

$$B_n = \frac{1}{L} \int_{L_g+H}^L f(\xi) \cos \frac{k\pi}{L} \xi d\xi \quad (30)$$

we get:

$$B_n = -\frac{1}{k\pi} \sin \frac{k\pi(L_g+H)}{L} \quad (31)$$

Similarly, the result for the transition zone is:

$$B_h = \frac{1}{k\pi} \left[\sin \frac{k\pi(L_g+H)}{L} - \sin \frac{k\pi L_g}{L} \right] \quad (32)$$

The expressions of each parameter in the equation of the condensate gas phase and transition zone are solved above. Substitute A , λ , B_n , and B_h into Equation (21), leads to:

$$f_n(t, x) = -\sum_{k=1}^{\infty} \frac{1}{k\pi} \sin \frac{k\pi(L_g+H)}{L} \exp\left(-\frac{k^2 \pi^2}{L^2} D_{ni} t\right) \cos \frac{k\pi}{L} x \quad (33)$$

$$f_h(t, x) = \sum_{k=1}^{\infty} \frac{1}{k\pi} \left[\sin \frac{k\pi(L_g+H)}{L} - \sin \frac{k\pi L_g}{L} \right] \exp\left(-\frac{k^2 \pi^2}{L^2} D_{ni} t\right) \cos \frac{k\pi}{L} x \quad (34)$$

Therefore, the concentration at different time and position is:

$$\frac{c_{ni}}{c_{ni}} = -\sum_{k=1}^{\infty} \frac{1}{k\pi} \sin \frac{k\pi(L_g+H)}{L} \exp\left(-\frac{k^2 \pi^2}{L^2} D_{ni} t\right) \cos \frac{k\pi}{L} x \quad (35)$$

$$\frac{c_{hi}}{c_{hi}} = \sum_{k=1}^{\infty} \frac{1}{k\pi} \left[\sin \frac{k\pi(L_g + H)}{L} - \sin \frac{k\pi L_g}{L} \right] \exp \left(-\frac{k^2 \pi^2}{L^2} D_{ni} t \right) \cos \frac{k\pi}{L} x \quad (36)$$

Assuming that the molar composition and molar concentration of component i are x_{hi} , and c_{hi} in the transition zone; the molar composition and molar concentration of component i in the condensate gas are x_{ni} , and c_{ni} ; the initial molar composition and molar concentration of component i in the transition zone are X_{hi} , and C_{hi} , the initial molar composition and molar concentration of component i in the condensate gas are X_{ni} , and C_{ni} , the molar concentration of the transition zone and condensate are c_h , and c_n , respectively. The deviation coefficient of the transition zone and condensate gas are Z_n , and Z_h , respectively. The initial deviation coefficient of the transition zone and condensate gas are Z_{h0} , Z_{n0} , and the initial pressure is P_0 . Considering the real gas state equation, we get:

$$x_{ni} = \frac{c_{ni}}{c_n} = \frac{c_{ni} Z_n R T}{P} \quad (37)$$

$$x_{hi} = \frac{c_{hi}}{c_h} = \frac{c_{hi} Z_h R T}{P} \quad (38)$$

$$C_{ni} = \frac{P_{n0} X_{ni}}{Z_{n0} R T} \quad (39)$$

$$C_{hi} = \frac{P_{h0} X_{hi}}{Z_{h0} R T} \quad (40)$$

The molar composition of component i in transition zone and condensate gas can be obtained by combining Equation (35) ~ (40):

$$x_{ni} = -\frac{P_0 Z_n X_{ni}}{Z_{n0} P} \sum_{k=1}^{\infty} \frac{1}{k\pi} \sin \frac{k\pi(L_g + H)}{L} \exp \left(-\frac{k^2 \pi^2}{L^2} D_{ni} t \right) \cos \frac{k\pi}{L} x \quad (41)$$

$$x_{hi} = \frac{P_0 Z_h X_{hi}}{Z_{h0} P} \sum_{k=1}^{\infty} \frac{1}{k\pi} \left[\sin \frac{k\pi(L_g + H)}{L} - \sin \frac{k\pi L_g}{L} \right] \exp \left(-\frac{k^2 \pi^2}{L^2} D_{ni} t \right) \cos \frac{k\pi}{L} x \quad (42)$$

Diffusion coefficients and EOS

The diffusion coefficient of component i in the gas phase mixture is calculated by the following equations:

$$D_{ig} = \frac{1 - x_{ig}}{\sum_{j \neq i} x_{ig} D_{ij}^{-1}} \quad (43)$$

$$D_{ij} = \frac{\rho_g^0 D_{ij}^0}{\rho_g} \left(0.99589 + 0.096016 \rho_{gr} - 0.22035 \rho_{gr}^2 + 0.032874 \rho_{gr}^3 \right) \quad (44)$$

$$\rho_{gr} = \rho_g \left(\frac{\sum_{i=1}^n x_{ig} v_{ci}^{5/3}}{\sum_{i=1}^n x_{ig} v_{ci}^{2/3}} \right) \quad (45)$$

$$\rho_g^0 D_{ij}^0 = \frac{1.8583 \times 10^{-3} T^{0.5}}{\sigma_{ij}^2 \Omega_{ij} R} \left(\frac{1}{M_i} + \frac{1}{M_j} \right)^{0.5} \quad (46)$$

Where σ is the potential constant, 10^{-10} m.

M_i is the molecular weight of component i .

$$\sigma_{ij} = (\sigma_i + \sigma_j) / 2;$$

Ω is diffusion collision integral, dimensionless.

$$\Omega_{ij} = \frac{1.06036}{T_N^{0.1561}} + \frac{0.193}{\exp(0.47635 T_N)} + \frac{1.03587}{\exp(1.52996 T_N)} + \frac{1.76474}{\exp(3.89411 T_N)} \quad (47)$$

$$T_N = KT / \varepsilon_{AB} \quad (48)$$

K is Boltzmann constant.

ε is potential energy constant.

The mixture deviation coefficient Z is calculated by DAK method (*Dranchuk, P.M., 1975*):

$$Z = 1 + (A_1 + A_2 T_{pr}^{-1} + A_3 T_{pr}^{-3} + A_4 T_{pr}^{-4} + A_5 T_{pr}^{-5}) \rho_{pr} + (A_6 + A_7 T_{pr}^{-1} + A_8 T_{pr}^{-2}) \rho_{pr}^2 - A_9 (A_7 T_{pr}^{-1} + A_8 T_{pr}^{-2}) \rho_{pr}^5 + A_{10} \left(1 + A_{11} \rho_{pr}^2 \right) \left(\rho_{pr}^2 T_{pr}^{-3} \right) e^{-A_{11} \rho_{pr}^2} \quad (49)$$

$$\rho_{pr} = \frac{0.27}{Z} \cdot \frac{P_{pr}}{T_{pr}} \quad (50)$$

Where, T_{pr} is the pseudo-contrast temperature, P_{pr} is the pseudo-contrast pressure, ρ_{pr} is the pseudo-contrast density; values of $A_1 \sim A_{11}$ are: $A_1=0.3265$, $A_2=-1.07$, $A_3=-0.5339$, $A_4=0.01569$, $A_5=-0.05165$, $A_6=0.5475$, $A_7=-0.7361$, $A_8=0.1844$, $A_9=0.1056$, $A_{10}=0.6134$, $A_{11}=0.721$.

Pressure P is calculated by PR equation (*Peng, D.Y., 1976*):

$$P = \frac{RT}{V - b_m} - \frac{a_m(T)}{V(V + b_m) + b_m(V - b_m)} \quad (51)$$

$$a_m(T) = \sum_{i=1}^n \sum_{j=1}^n x_i x_j (a_i a_j \alpha_i \alpha_j)^{0.5} (1 - k_{ij}) \quad (52)$$

$$b_m = \sum_{i=1}^n x_i b_i \quad (53)$$

$$a_i = 0.45724 \frac{R^2 T_{ci}}{P_{ci}} \quad (54)$$

$$b_i = 0.0778 \frac{RT_{ci}}{P_{ci}} \quad (55)$$

$$\alpha_i = [1 + m_i (1 - T_{ri}^{0.5})]^2 \quad (56)$$

$$T_{ri} = \frac{T}{T_{ci}} \quad (57)$$

$$m_i = 0.3464 + 1.5426 \omega_i - 0.26992 \omega_i \quad (58)$$

Where k_{ij} is the binary interaction coefficient; ω_i is the acentric factor of component i .

Results and discussion

Static parameters

In order to run the subsequent dynamic parameter calculation and analysis, the first step is to calculate the initial static parameters. The deviation coefficient of dry gas is 1.1893; the deviation coefficient of condensate gas (with high oil content) is 1.1912, slightly higher than that of injected dry gas. The density of injected gas is 204.76 kg/m³, and the density of condensate gas is 281.70 kg/m³.

The difference between the two gases is substantial, leading to the formation of a visible interface in the early stage. Table 2 shows the calculated diffusion coefficients. The results show that the molecular motion ability is weakened in the condensate gas due to a large number of heavier components, larger molecular weight, and stronger intermolecular force. Thus, the diffusion coefficient value of each component in the condensate gas is 1-2 orders of magnitude lower than that in the dry gas. Moreover, the diffusion coefficients of components in the condensate gas are also significantly different, with C1 being the largest, and the diffusion coefficients decrease by orders of magnitude with the increase of molecular weight.

Dynamic diffusion coefficient

The dynamic diffusion coefficient of each component in the diffusion process is calculated based on the calculation of static parameters; Figure 5, Figure 6, and Figure 7 show that the diffusion coefficient of each component decreases gradually with the diffusion process. At the initial diffusion stage, molecular diffusion is more intense due to the significant concentration difference between the dry gas and condensate gas, leading to a rapid decrease in the composition and concentration difference between the two phases. Thus, the diffusion coefficient decreases quickly at the early stage. However, the change in the diffusion coefficient gradually becomes stable in the later stage of diffusion.

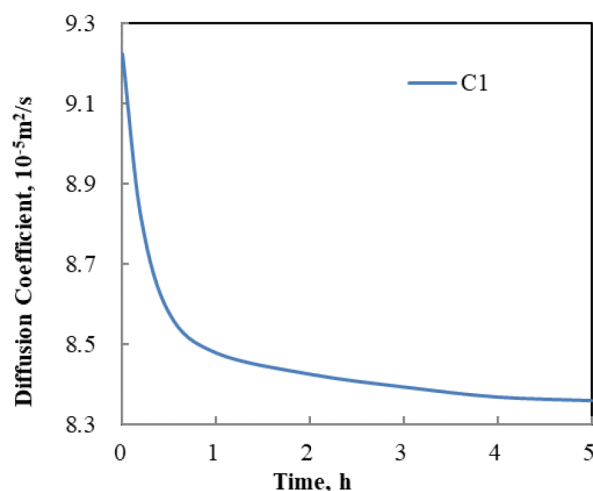


Figure 5—Curve of diffusion coefficient C1 with time

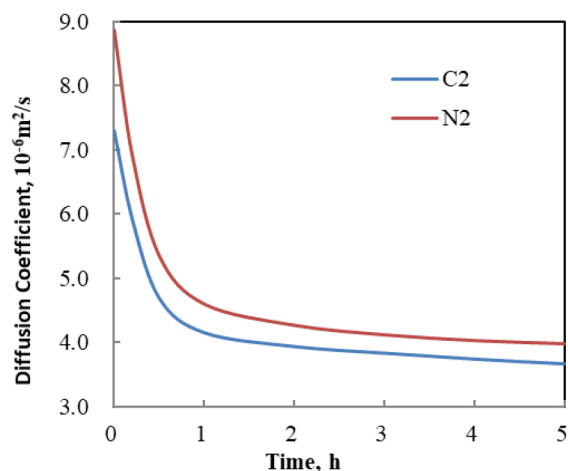


Figure 6—Curve of diffusion coefficient C2, N2 with time

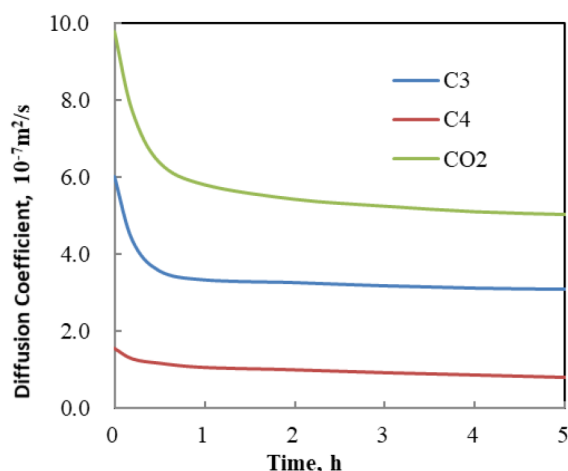


Figure 7—Curve of diffusion coefficient C3, C4, CO2 with time

Density and Diffusivity Trends

The development of condensate reserves often accompanies the accumulation of condensate oil, and it is essential to understand the changes of some critical parameters with condensate oil.

Figure 8 shows the density change at different timing and different positions in the PVT tube. The density difference between the lower condensate gas and the upper dry gas is evident at the early diffusion stage. With the diffusion process going on, the upper gas density gradually increases, and the lower gas density gradually decreases. Namely, the density difference between the upper and lower parts gradually decreases, and the density curve gradually approaches the intermediate value.

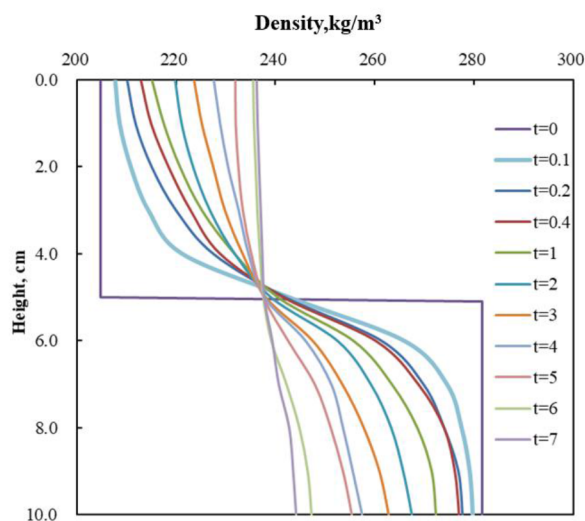


Figure 8—Density changes during the diffusion process (PVT tube)

Besides, the system's pressure reduces due to the molecular diffusion effect, and the average density of the system is lower than that at the initial state. It can also be observed that the density mutation zone in the middle gradually increases in thickness, indicating that the transition zone is growing. In the later stage, the upper and lower densities gradually equalize. After 3 hours, the density gradient becomes smaller, and an obvious interface can barely be observed, implying that the whole system is in an equilibrium state.

To better understand the diffusion process at the field scale, we reset the geometry of the studied area. Specifically, the overall formation thickness is 100 m with the injected gas 50 m and the condensate gas 50

m. In addition, as one noticeable feature in condensate gas development is the occurrence of the condensate oil, we calculated diffusion coefficients in condensate gas with different oil contents.

Figure 9 shows that the density difference between the top and bottom is around 14 kg/m^3 after 800 hours, and we assume that an equilibrium state is achieved at that condition. However, it only takes about 500 hours to reach the same condition for the system with less condensate oil (Figure 10). Generally, a system with lower oil content is more likely to achieve an equilibrium state during a short period.

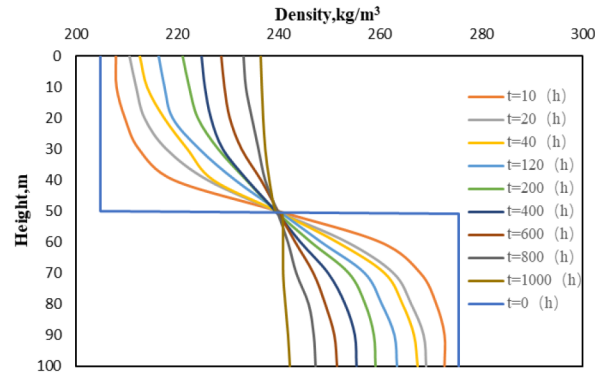


Figure 9—Density changes during the diffusion process (more oil content)

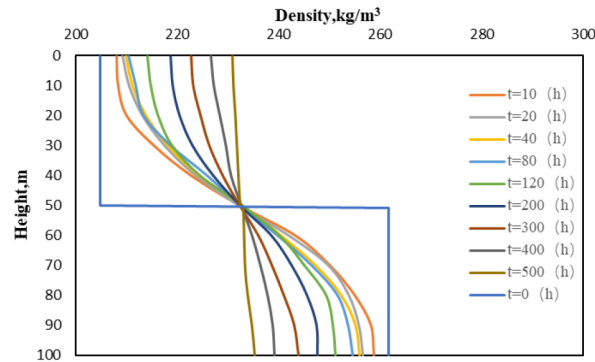


Figure 10—Density changes during the diffusion process (less oil content)

It also can be observed (Figure 11 to Figure 13) that diffusion coefficients of the lighter systems are more prominent at the initial period, indicating a more active diffusion process. At this period, the system with a lighter molecular weight dominates the diffusion process. Later, the concentration difference between injected gas and condensate gas with less oil content is lower than that between injected gas and condensate gas with high oil content, and high concentration difference may play a vital role in diffusion activity at that time. This helps explain the drop in the diffusion coefficient at a later stage for the system with lower oil content.

The results also illustrate the phenomenon shown in Figure 2 and Figure 3 that it takes longer for the high oil content system to observe the disappearance of the interface.

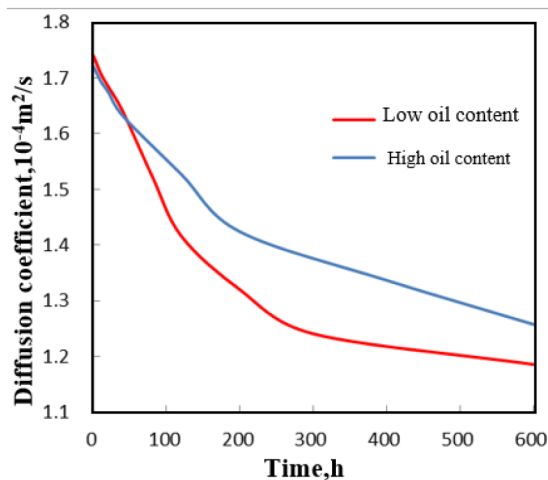


Figure 11—Diffusion coefficient of C1

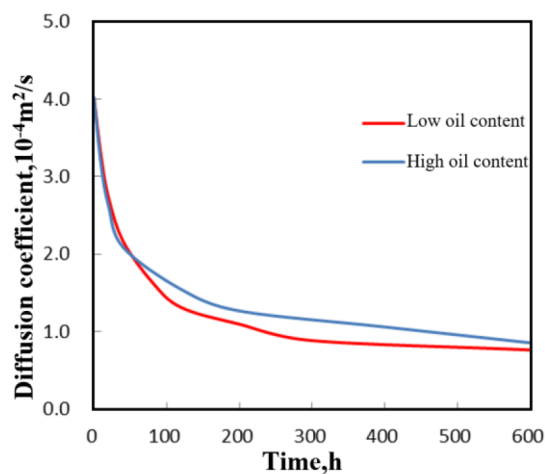


Figure 12—Diffusion coefficient of C2

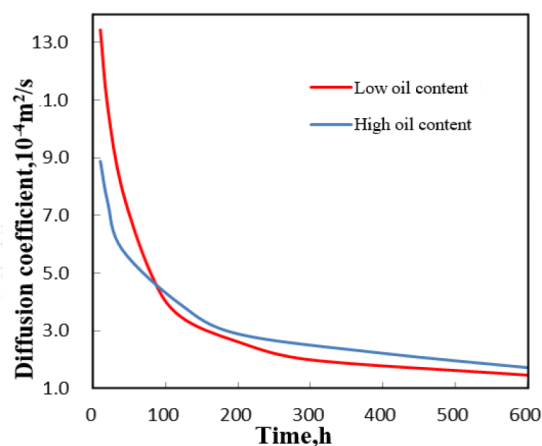


Figure 13—Diffusion coefficient of C7+

Conclusions

The visual PVT experiment demonstrates the vertical diffusion process between condensate gas and dry gas at a non-equilibrium state, providing the experimental basis for the theoretical model.

A non-equilibrium diffusion model of supercritical condensate gas and dry gas under high temperature and pressure conditions is established. The model considers the interaction between various components, which shows excellent agreement to the actual situation.

Under reservoir conditions, the density difference between supercritical condensate gas and dry gas injected is obvious. The diffusion coefficient of different components varies from 10^{-7} m²/s to 10^{-5} m²/s. In comparison, the conventional values are around 10^{-5} m²/s to 10^{-4} m²/s. Therefore, introducing a dynamic diffusion coefficient is of substantial practical value for understanding and evaluating the gas injection properties.

In the diffusion process, the diffusion coefficient decreases rapidly at the initial stage and stabilizes at the later stage. The diffusion coefficient becomes smaller with the increase of molecular weight; The density difference between dry gas and condensate gas decreases gradually, accompanied by the thickening of the transition zone till the two phases reach an equilibrium state.

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Nomenclature

x	the distance from the top of the PVT cell, cm
L	total height of the system, cm
L_g	the height of the sections of dry gas L_g , cm
L_n	the height of the sections of condensate gas L_n , cm
H	thickness of transition zone, cm
t	the diffusion time, h
c	the molar concentration, mol/cm ³
c_{hi}	the molar concentration of component i in the transition zone, mol/cm ³
c_{ni}	the molar concentration of component i in condensate gas, mol/cm ³
C_{ni}	initial molar concentration of component i in condensate gas, mol/cm ³
C_{hi}	initial molar concentration of component i in the transition zone, mol/cm ³
D_{hi}	diffusion coefficients of components i in transition zone, cm ² /s
f	dimensionless concentration of condensate gas
h	the function of dimensionless concentration with respect to x
g	the function of dimensionless concentration with respect to t
K, K_1, K_2, A, B, B_h	coefficient of general solution
a	the constant
λ	the coefficient, cm ⁻¹
ξ	the intermediate variable
f_n	dimensionless concentration of condensate gas
f_h	dimensionless concentration of transition zone
i	component number
x_{hi}	the molar composition of component i in the transition zone
x_{ni}	the molar composition of component i in condensate gas
X_{hi}	the initial molar composition of component i in the transition zone
X_{ni}	the initial molar composition of component i in condensate gas
ch	the molar concentration of transition zone, mol/cm ³
c_n	the molar concentration of condensate gas, mol/cm ³
Z_n	the compressibility factor of condensate gas

Z_h	the compressibility factor of transition zone
Z_{h0}	the initial compressibility factor of transition zone
P_0	initial pressure, MPa
D_{ig}	the diffusion coefficient of component i in gas phase, cm^2/s
x_{ig}	the molar composition of component i in gas phase
ρ_g	the gas density, g/cm^3
D_{ij}	interdiffusion coefficient of component i and component j , cm^2/s
M_i	the molecular weight of component i
M_j	the molecular weight of component j
R	the molar gas constant
K	Temperature
$\sigma_{ij}\sigma_i\sigma_j$	the potential constant, 10^{-10}m
Ω_{ij}	the diffusion collision integral
ε_{AB}	the potential energy constant
Z	the compressibility factor
T_{pr}	the pseudo-contrast temperature
P_{pr}	the pseudo-contrast pressure
k_{ij}	the binary interaction coefficient
ω_i	the acentric factor of component i

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